HW3- Computer Simulation II

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1

Using the code from the first homework (where I already had implemented the Wolff algorithm for improvements in computational time) we can easly calculate $dU/d\beta$ at β_c:

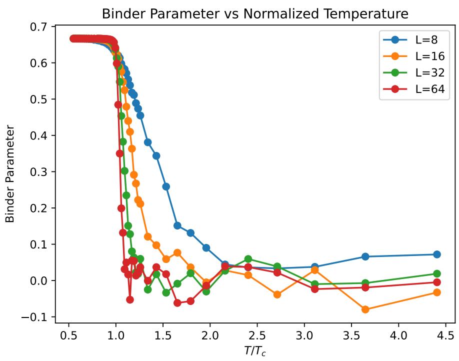


Figure 1: Binder parameter

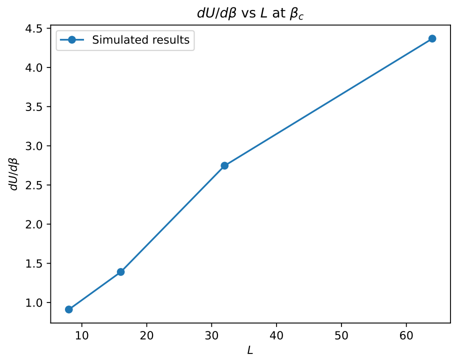
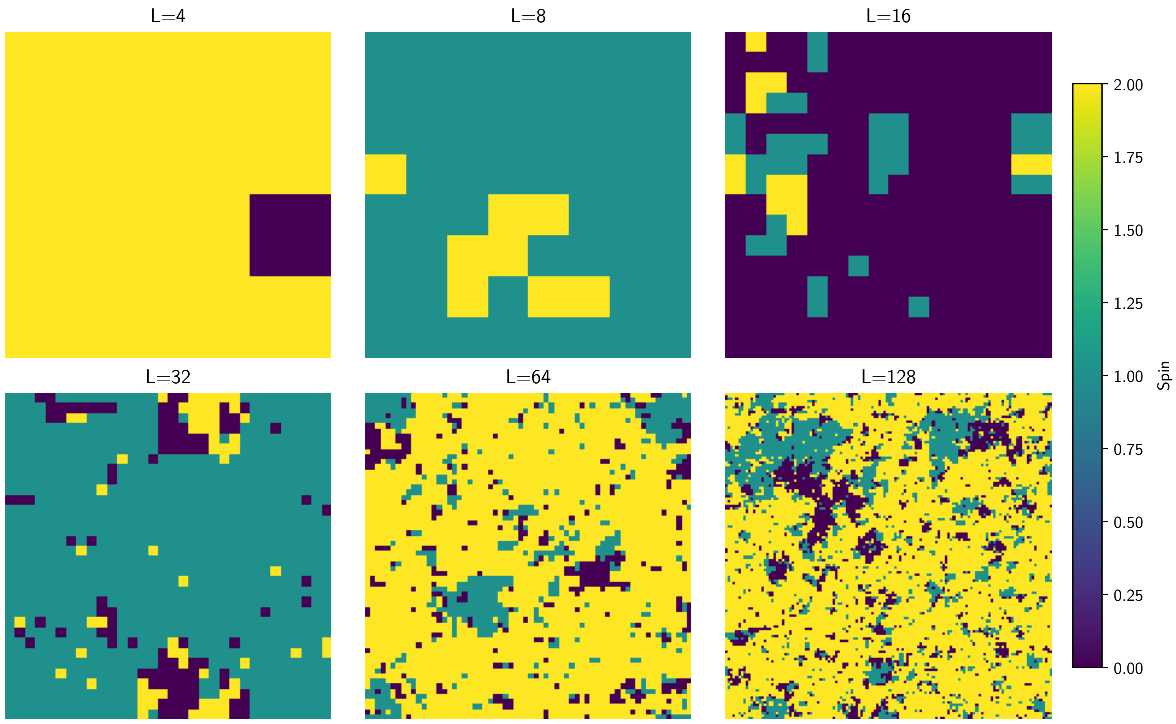


Figure 2: derivative of the Binder parameter

We can see that $dU/d\beta$ scales linearly with lattice size.

2

Using the q-states Potts model and $q = 3$, $\beta = \ln(1 + \sqrt{q})$, we can analyze the behavior of the system for different lattice sizes L. The following graph illustrate the equilibrium state of the system for different lattice sizes L:



We can see the distinc phases that arrise at the critical temperature, we can further calculate the Energy and Specific heat of the lattices. Plotting it in function of lattice size L:

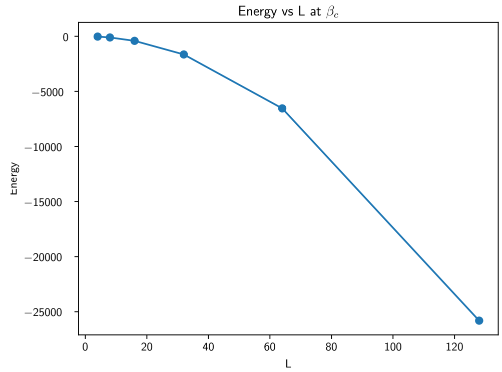


Figure 3: Energy

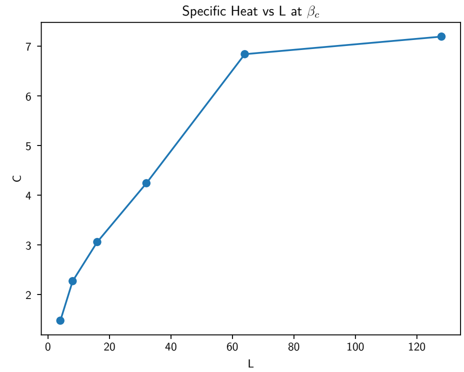


Figure 4: Specific Heat

We can see that the energy and specific heat seem to be dependent on powers of L, lets see how this graphs look in a log-scale:

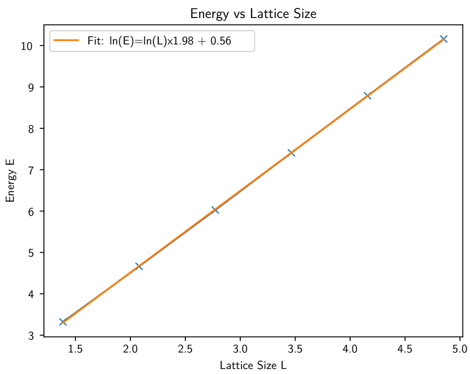


Figure 5: Absolute Energy

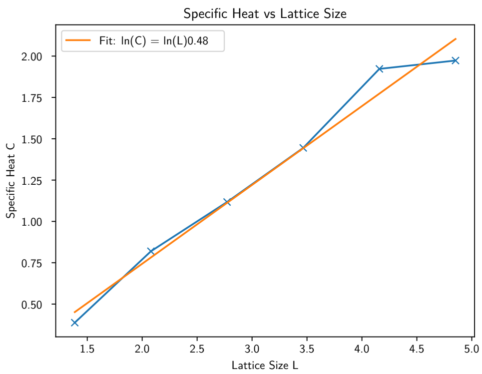


Figure 6: Specific Heat

So we see the relations:

$$E \propto -L^{1.98} \quad \text{and} \quad C \propto L^{0.48}$$